



PROJECT REPORT

Dual-Axis Solar Tracking and Energy Management System

1. Primary Objectives

Maximize Energy Capture

To dynamically track the sun's trajectory in real-time across two degrees of freedom (Yaw and Pitch), minimizing the angle of incidence to maximize the solar panel's photovoltaic efficiency.

Environmental Telemetry

To integrate embedded ambient sensors to log real-time microclimate metrics (temperature and humidity) that correlate with solar output degradation.

Closed-Loop Power Monitoring

To accurately log electronic metrics including open-circuit voltage, load current, and instantaneous power dissipation using high-precision I²C sensor hardware.

Efficient Load Regulation & Storage

To build a safe energy-storage system utilizing dedicated battery charging profiles to prevent voltage collapse, reverse-current leakage, and thermal runaway.

2. Hardware Components

VISUAL	COMPONENT NAME	TECHNICAL SPECIFICATIONS	PURPOSE IN PROJECT
	Arduino Uno R3	ATmega328P, 5V Logic	Central processing unit for sensor computation and PWM actuation.
	Solar PV Panel	5V - 6V, 200mA - 500mA	Primary transducer converting solar irradiance into electrical energy.
	LDRs (×4)	10kΩ Photoresistors	Arranged in a cross-axis quadrant to sense light differential.
	MG996R Servos	Metal Gear, 4.8V–7.2V	Actuators for executing Horizontal (Yaw) and Vertical (Pitch) rotation.
	DHT22 Sensor	Capacitive Humidity/Temp	Measures ambient environmental temperature and relative humidity.
	INA219 Breakout	12-bit ADC, I²C	High-side current and bus voltage monitoring module.
	TP4056 & LM2596	Charger & Buck Converter	Safeguards storage cells and steps down voltage for servos.
	LiPo & Li-Ion Cells	7.4V (2S) & 3.7V Storage	Power source for servos and storage medium for solar energy.

3. System Schematic Diagram

[illegible]

4. Integrated Production Code

```
#include <Servo.h>
#include <DHT.h>
#include <Wire.h>
#include <Adafruit_INA219.h>

#define DHTPIN 2
#define DHTTYPE DHT22

// Transducer Interface Pins
const int pinTop    = A0;
const int pinBottom = A1;
const int pinLeft   = A2;
const int pinRight  = A3;

// Actuator Driving Pins
const int servoH_Pin = 9;
const int servoV_Pin = 10;

// Dynamic Control Variables
const bool INVERT_VERTICAL    = false;
const bool INVERT_HORIZONTAL = false;
const int tolerance = 40;    // Angular deadzone filter
const int stepSize = 1;      // Intermittent motor angular step

const int limitH_Min = 10;  const int limitH_Max = 170;
const int limitV_Min = 20;  const int limitV_Max = 160;

DHT dht(DHTPIN, DHTTYPE);
Adafruit_INA219 ina219;
Servo servoH; Servo servoV;

int posH = 90; int posV = 90;
unsigned long prevMillis = 0;
const long telemetryInterval = 2000;

void setup() {
  Serial.begin(9600);
  servoH.attach(servoH_Pin);
  servoV.attach(servoV_Pin);
  servoH.write(posH);
```

```

servoV.write(posV);

dht.begin();

if (!ina219.begin()) {
    Serial.println(F("[FATAL ERROR]: INA219 Comm Breakdown."));
    while (1) { delay(10); }
}
}

void loop() {
    // Capture Light Gradients via Voltage Dividers
    int top    = 1023 - analogRead(pinTop);
    int bottom = 1023 - analogRead(pinBottom);
    int left   = 1023 - analogRead(pinLeft);
    int right  = 1023 - analogRead(pinRight);

    // Mathematical Angular Correction Logic
    int diffV = top - bottom;
    if (abs(diffV) > tolerance) {
        if (top > bottom) posV += (INVERT_VERTICAL ? -stepSize : stepSize);
        else posV += (INVERT_VERTICAL ? stepSize : -stepSize);
    }

    int diffH = left - right;
    if (abs(diffH) > tolerance) {
        if (left > right) posH += (INVERT_HORIZONTAL ? stepSize : -stepSize);
        else posH += (INVERT_HORIZONTAL ? -stepSize : stepSize);
    }

    // Enforce Boundary Conditions to Prevent Servo Stalling
    posV = constrain(posV, limitV_Min, limitV_Max);
    posH = constrain(posH, limitH_Min, limitH_Max);

    servoV.write(posV);
    servoH.write(posH);

    // Non-Blocking Execution of System Telemetry
    unsigned long currentMillis = millis();
    if (currentMillis - prevMillis >= telemetryInterval) {
        prevMillis = currentMillis;

        float humidity = dht.readHumidity();
        float temperature = dht.readTemperature();
        float busVoltage = ina219.getBusVoltage_V();
    }
}

```

```

float current_mA = ina219.getCurrent_mA();
float power_mW   = ina219.getPower_mW();

Serial.println(F("
>>> METRIC TELEMETRY LOG <<<"));
Serial.print(F("Temp: ")); Serial.print(temperature, 1);
Serial.print(F(" °C | Hum: ")); Serial.print(humidity, 1); Serial.println(F(" %"));
Serial.print(F("Bus: ")); Serial.print(busVoltage, 2); Serial.print(F(" V"));
Serial.print(F(" | Current: ")); Serial.print(current_mA, 1); Serial.print(F("
mA"));
Serial.println(F(""));
}
delay(30);
}

```

5. Future Scope

- **IoT Cloud Data Analytics:** Implementing an ESP32 or Wi-Fi gateway module to stream the local ambient data and electrical logs onto cloud servers (like ThingSpeak or AWS IoT) for computational analysis on solar degradation curves over time.
- **Maximum Power Point Tracking (MPPT):** Replacing the standard linear charging module with a custom microcontroller-driven buck-boost MPPT topology to maximize solar output efficiency during heavy clouds or atmospheric shading.
- **Autonomous Self-Powering Architecture:** Restructuring the energy paths such that the stored solar energy cycles through a secondary boost rail to feed power back into the microcontrollers, making the setup completely independent of external grid charging.

6. Conclusion

The designed prototype successfully presents a working framework for a dual-axis solar tracking and real-time analytical instrument. By using a quadrant-based LDR sensor module, the physical system aligns itself to the brightest optical spot with high accuracy.

The integration of isolated high-current rails using buck regulators prevents micro-controller brownouts, while the high-side INA219 instrumentation telemetry accurately quantifies load

profiles without disrupting circuit parameters. The implementation of strict safe-charging protocols shields the lithium storage cells from over-voltage stress.

Ultimately, this project confirms that active dual-axis tracking significantly enhances power harvest limits compared to traditional stationary structures, presenting a viable baseline model for modern green energy systems.